Math Kangaroo Practice

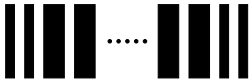
Grade 3-4 · 20 questions · harder · seed 11572

Part A: Each question is worth 3 points

1. How many different letters are used to write the word WINDOW?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

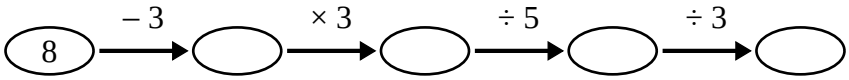
2. A bar code is made of 21 black and white bars, black and white taking turns. The first and the last bar are black. Each black bar is either wide or narrow. There are 7 more white bars than wide black bars.



How many narrow black bars are there?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 4 ***E.*** 3

3. Follow the arrows and do each calculation in turn.

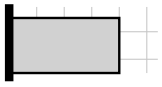


Which number comes out at the end?

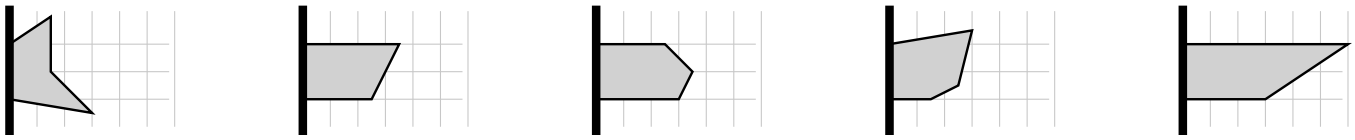
A. 8 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

Part B: Each question is worth 4 points

4. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Tom draws the shadow at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



A B C D E

How many of the five drawings can be right?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

Part C: Each question is worth 5 points

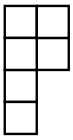
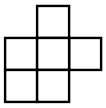
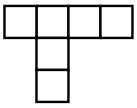
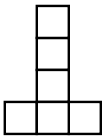
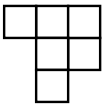
5. Anna and Lucy are twins. Anna was born at 11:48 p.m. on December 31, 2006. Lucy was born 50 minutes later.
What is Lucy's date of birth?

A. December 31, 2007 ***B.*** January 1, 2005 ***C.*** January 1, 2007
D. December 31, 2006 ***E.*** January 2, 2007

6. Five children, Ben, David, Iris, Felix and Grace, sit in a row of 5 chairs at the cinema. Ben and David are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 24 ***B.*** 48 ***C.*** 50 ***D.*** 72 ***E.*** 120

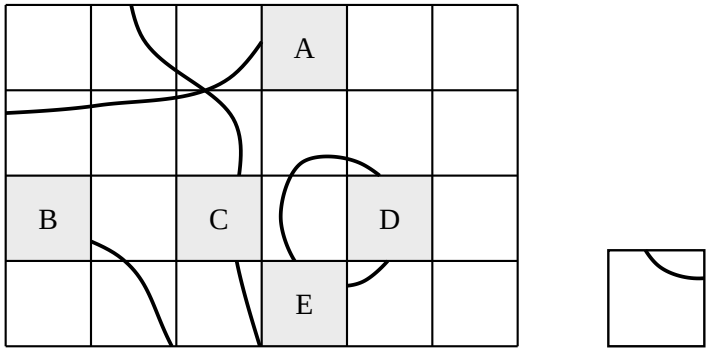
7. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

8. Finn's new rug is 4 m 5 cm long. Four of these lengths are the same as that.
Which one is **not** the same?

A. 45 dm ***B.*** 4 m 50 mm ***C.*** 405 cm ***D.*** 40 dm 5 cm ***E.*** 4050 mm

9. A cat played with balls of wool on squared paper. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

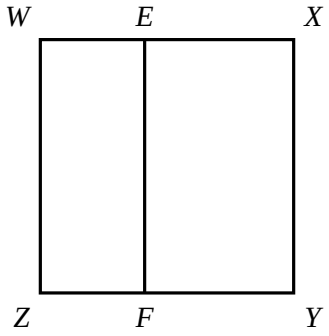
10. The odometer of a scooter shows 2469 km. Each digit is bigger than the digit before it.
After how many more kilometres will this happen again?

A. 19 ***B.*** 9 ***C.*** 8 ***D.*** 7 ***E.*** 1

11. How many three-digit numbers have digits that add up to 7?

A. 56 ***B.*** 35 ***C.*** 34 ***D.*** 29 ***E.*** 28

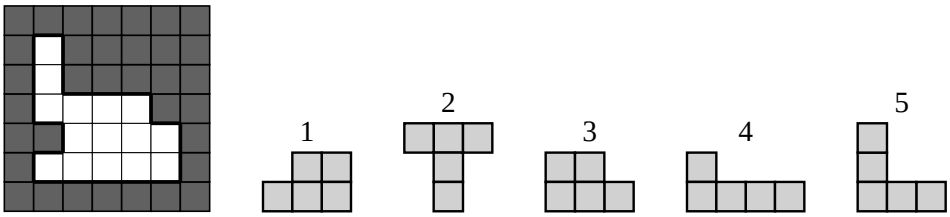
12. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 48 cm and the perimeter of *EXYF* is 54 cm.



How long is the side of the square *WXYZ*?

A. 6 cm ***B.*** 13 cm ***C.*** 15 cm ***D.*** 16 cm ***E.*** 17 cm

13. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



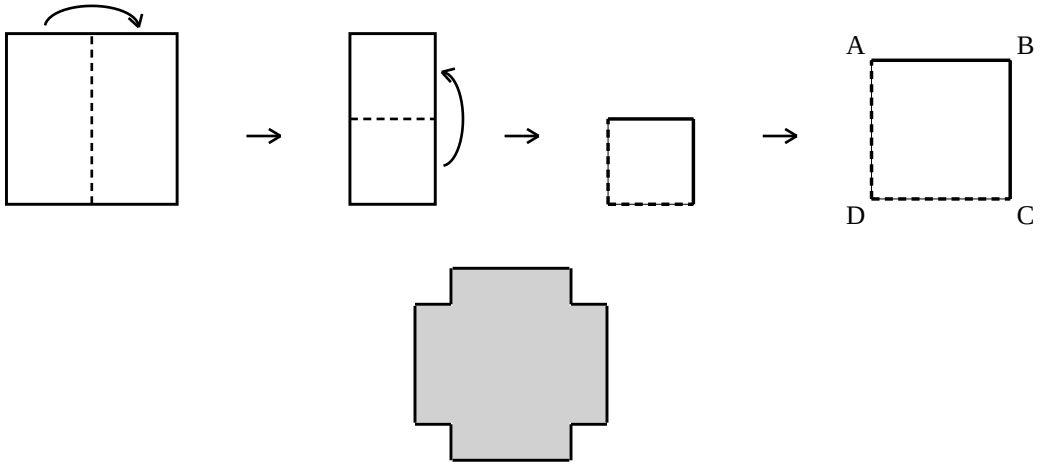
A. 1+2+3 ***B.*** 1+3+5 ***C.*** 2+3+4 ***D.*** 2+3+5 ***E.*** 3+4+5

14. A big cube made of small cubes is painted on the outside and taken apart. Exactly 36 small cubes have 2 painted faces.

How many small cubes long is each edge of the big cube?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

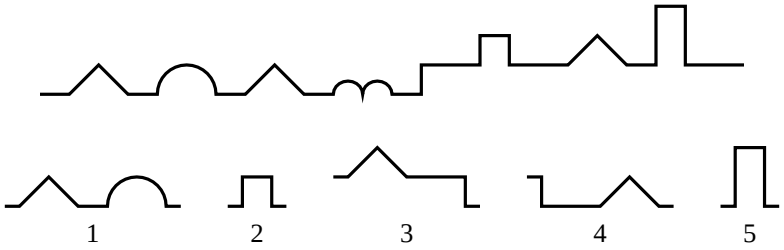
15. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square was cut off, and after unfolding the paper looks like the bottom picture.



Which corner was cut off?

A. corner A ***B.*** corner B ***C.*** corner C ***D.*** corner D ***E.*** it cannot be made with one cut

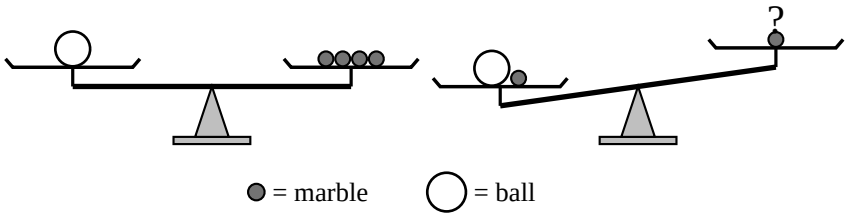
16. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

17. The first balance is level. On the second balance there is 1 ball and 1 marble on the left and 1 marble on the right.



How many marbles must be added to the right plate to make it level?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 6

18. Tom, Julien, Ravi, Omar and Emma each have a different pet: a turtle, a cat, a parrot, a dog and a goldfish. Julien and Omar both have a furry animal. Emma has a pet with a shell. Ravi has a pet that lives in water. Omar has a pet that barks.
Which statement is **not** true?

A. Tom has a parrot ***B.*** Emma has a turtle ***C.*** Tom has a turtle
D. Ravi has a goldfish ***E.*** Omar has a dog

19. Mia types the numbers from 1 to 150 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,150.
How many keys does Mia press (each digit and each comma is one key)?

A. 491 ***B.*** 492 ***C.*** 495 ***D.*** 500 ***E.*** 566

20. Leo has the cards 1, 2, 5 and 8. Using every card exactly once, with +, –, ×, ÷ and brackets, Leo wants to get the biggest possible number.
What is the biggest number Leo can get?

A. 80 ***B.*** 120 ***C.*** 136 ***D.*** 143 ***E.*** 147

Answer Key

1.	**D**	**2.**	**A**	**3.**	**E**	**4.**	**E**	**5.**	**C**	**6.**	**B**
7.	**A**	**8.**	**A**	**9.**	**E**	**10.**	**B**	**11.**	**E**	**12.**	**E**
13.	**B**	**14.**	**A**	**15.**	**B**	**16.**	**D**	**17.**	**D**	**18.**	**C**
19.	**A**	**20.**	**B**								

Solutions

1. D (5) — The letters are D, I, N, O and W: 5 different letters (WINDOW has 6 letters in all). [2007 Q3 · different_letters]

2. A (8) — Black is at both ends, so there are 11 black and 10 white bars. Wide: 10 – 7 = 3. Narrow: 11 – 3 = 8. [2003 Q17 · narrow_bars]

3. E (1) — 8 – 3 = 5, 5 × 3 = 15, 15 ÷ 5 = 3, 3 ÷ 3 = 1. [2007 Q5 · forwards]

4. E (4) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: E. So 4 can be right. [2002 Q15 · count_possible]

5. C (January 1, 2007) — 12 minutes after 11:48 p.m. it is midnight, and a new day and a new year start. Lucy was born 38 minutes after midnight, on January 1, 2007. So Anna is older, but they have different birthdays and were even born in different years. [2007 Q12 · new_year_twins]

6. B (48) — Think of Ben and David as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block Ben and David can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

7. A — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece A fit together like that. [2007 Q8 · two_copies]

8. A (45 dm) — 4 m 5 cm = 405 cm = 4050 mm. But 45 dm = 450 cm. [2003 Q10 · not_equal]

9. E (gap E) — The square has top: 1 line, right: 1 line, bottom: no line, left: no line. Only gap E has lines reaching it at exactly those places. [2002 Q1 · which_gap]

10. B (9) — Keep as many digits on the left as possible. The next such number is 2478: 2478 – 2469 = 9. [2002 Q24 · going_up]

11. E (28) — Count by the first digit: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28. [2007 Q4 · count_with_sum]

12. E (17 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 48 + 54 = 102 cm = 6 sides. Side = 102 ÷ 6 = 17 cm. [2002 Q6 · two_perimeters]

13. B (1+3+5) — The empty area has 15 squares, so the three sizes must add up to 15. Only 1 + 3 + 5 also fit together in the right shape. [2003 Q18 · three_pieces]

14. A (5) — A cube has 12 edges, so each edge has 36 ÷ 12 = 3 cubes between its corners. With the 2 corners, an edge is 5 cubes long. [2007 Q17 · how_big]

15. B (corner B) — The design shows all four corners cut off. That comes from cutting corner B (the real paper corner, away from the folds). [2007 Q24 · which_corner]

16. D (4) — Pieces 1, 2, 4, 5 fit. Not: 3 (it has the right shapes but in the wrong order). [2002 Q7 · how_many_fit]

17. D (4) — One ball = 4 marbles, so the left plate is worth 1 × 4 + 1 = 5 marbles. The right plate has 1, so add 5 – 1 = 4. [2002 Q9 · what_to_add]

18. C (Tom has a turtle) — The only possible matching: Tom – parrot; Julien – cat; Ravi – goldfish; Omar – dog; Emma – turtle. [2002 Q18 · which_not_true]

19. A (491) — Digits: 9 × 1 + 90 × 2 + 51 × 3 = 342. Commas: one fewer than the numbers, 149. 342 + 149 = 491. [2007 Q23 · with_commas]

20. B (120) — (1 + 2) × 5 × 8 = 120. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]